# Resampling

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Multivariate Statistics for Applied Data Science (ADSC 4050)

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Winter 2026

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## Topics

# Recap

- Last time we covered **Boosting II**:
  - Stochastic GBMs.
  - xgboost.
  - Hyperparameter tuning.

## Introduction

- We have already covered methods of resampling and their applications.
    - Bootstrap aggregation is a resampling method.
    - Hyperparameter tuning in random forests and GBMs are also examples of resampling.

- Resampling approaches can also be used to deal with class imbalances.

## Class Imbalance

- A **class imbalance** occurs when a dataset has an unequal distribution of classes.
  - Typically a single (majority) class heavily *dominates* the other (minority) class.
- It can become very difficult to predict the minority class.
  - Examples: Fraud detection, disease detection, and quality control.
- *Our classification models may be very accurate, but do not predict the minority class well.*

## Example 1

- Import the *GermanCredit* dataset from the caret package and do the following:
  1. Take some time to get to know the dataset.
  2. Examine the `Class` variable and see if there is an imbalance.
  3. Use the provided code to conduct a 80/20 train/test split.
  4. Perform cross validation on an out-of-the-box random forest to classify the credit class.
  5. Evaluate the overall performance and the performance on the minority class in the cross validation process and on the testing set.
- What do you notice (Accuracy, Sensitivity, and Specificity)?

## Resampling Solutions

- There are steps that we can take to help our models classify when we have class imbalances:

  1. **Downsampling:** Randomly subset all the classes in the training set so that their class frequencies match the least prevalent class.
  2. **Upsampling:** Randomly sample (with replacement) the minority class to be the same size as the majority class.
  3. **Hybrid methods:** Down-sample the majority class and synthesize new data points in the minority class (SMOTE & ROSE).

- **You must not resample your validation or testing sets!**

# (1) Downsampling

- **Downsampling** is the process of randomly subset all the classes in the training set so that their class frequencies match the least prevalent class.

- Reduces the overall size of your training data.

- Using the caret package in R:

```
down_train <- downSample(
x = train_data[, -ncol(train_data)],
y = train_data$Class)
```

  - We can also add the `sampling = "down"` argument to the `trainControl()` function.

## Example 2

- Using the *GermanCredit* dataset from the caret package, do the following:
    1. Downsample the dataset to obtain a new training set with a balanced class.
    2. Perform cross validation on an out-of-the-box random forest to classify the credit class.
    3. Evaluate the overall performance and the performance on the minority class in the cross validation process and on the testing set.
- What do you notice (Accuracy, Sensitivity, and Specificity)?

# Upsampling

- **Upsampling** is the process of randomly sampling (with replacement) the minority class to be the same size as the majority class.

- Increases the overall size of your training data.
- Using the caret package in R:

```
up_train <- upSample(
x = train_data[, -ncol(train_data)],
y = train_data$Class)
```

  - We can also add the `sampling = "up"` argument to the `trainControl()` function.

## Example 3

- Using the *GermanCredit* dataset from the caret package, do the following:
  1. Upsample the dataset to obtain a new training set with a balanced class.
  2. Perform cross validation on an out-of-the-box random forest to classify the credit class.
  3. Evaluate the overall performance and the performance on the minority class in the cross validation process and on the testing set.
- What do you notice (Accuracy, Sensitivity, and Specificity)?

# Hybrid Methods

## Hybrid Methods

- **Hybrid methods** typically down-sample the majority class and synthesize new data points in the minority class.

- We will cover two hybrid methods in this course:
  1. **Synthetic minority over-sampling technique** (SMOTE): Creates new synthetic examples for the minority class by interpolating between existing minority samples and their nearest neighbours.
  2. **Random over-sampling examples** (ROSE): Generates synthetic, plausible data examples based on conditional density estimates to create a balanced training set.

- Note: *Both hybrid methods may struggle when there are many noisy variables.*

## SMOTE Resampling in R

- SMOTE does not upsample directly, but creates similar observations to those in the minority class.
- We can use the themis package in R:

```
rec <- recipe(Class ~ ., data = train_data) %>%
 step_smote(Class)
rec_prep <- prep(rec)
smote_train <- bake(rec_prep, new_data = NULL)
```

## Example 4

- Using the *GermanCredit* dataset from the caret package, do the following:
    1. Apply SMOTE to the dataset to obtain a new training set with a balanced class.
    2. Perform cross validation on an out-of-the-box random forest to classify the credit class.
    3. Evaluate the overall performance and the performance on the minority class in the cross validation process and on the testing set.
- What do you notice (Accuracy, Sensitivity, and Specificity)?

## ROSE in R

- ROSE creates similar observations to the ones in the minority class based on a conditional density estimate.
- We can use the ROSE package in R:

```
rose_train <- ROSE(Class ~ ., data = train_data)$data
```

## Example 5

- Using the *GermanCredit* dataset from the caret package, do the following:
  1. Apply ROSE to the dataset to obtain a new training set with a balanced class.
  2. Perform cross validation on an out-of-the-box random forest to classify the credit class.
  3. Evaluate the overall performance and the performance on the minority class in the cross validation process and on the testing set.
- What do you notice (Accuracy, Sensitivity, and Specificity)?

## Final Thoughts

- These methods can be useful in instances with class imbalances.
  - The overall accuracy may decrease, but the *sensitivity* should increase.
- **Do not apply any of these methods to you validation or testing datasets!**
- What we have covered is not exhaustive, there are other methods:
  - ADASYN (Adaptive Synthetic Sampling).
  - Tomek Links.
  - NearMiss.

## Exercise 1

- Import the *Default* dataset from the ISLR package and do the following:
    1. Choose a binary classifier and try to predict the default class.
    2. Apply the resampling approaches that we covered to the dataset to obtain a new training set with a balanced class.
    3. Perform cross validation on your resampled models to classify the default class.
    4. Evaluate the overall performance and the performance on the minority class in the cross validation process and on the testing set.
    5. Which approach worked best?

## References & Resources


1. Bakker, J. D. (2026) Applied Multivariate Statistics in R
2. Peng R. D. (2020) *Exploratory Data Analysis with R*
3. Timbers T., Campbell T., and Lee M. (2024) *Data science: A first introduction.* Chapman and Hall/CRC
4. Bradley Boehmke B., and Greenwell B. (2020) *Hands-On Machine Learning with R.* Taylor & Francis


- Class Imbalances
- caret
- ROSE
- themis